

FREEDOM INTERNATIONAL SCHOOL

WORKSHEET- MCQ (SOLUTIONS)

PHYSICS

THERMAL PROPERTIES OF MATTER

CLASS XI

1. A composite rod made of copper ($\alpha = 1.8 \times 10^{-5} \text{ K}^{-1}$) and steel ($\alpha = 1.2 \times 10^{-5} \text{ K}^{-1}$) is heated. Then
- (a) it bends with steel on concave side (b) it bends with copper on concave side
(c) it does not expand (d) data is insufficient

Ans: (a) it bends with steel on concave side

Copper expands more than steel. So rod bends with copper on convex side and steel on concave side.

2. If the temperature of a rod is increased by 10°C , its length increases by 1%. What is the percentage change in volume of the body of the same material for 10°C increase in temperature?
- (a) 9% (b) 1% (c) 5% (d) 3%

Ans: (d) 3%

$$\Delta T = 10^\circ\text{C}$$

$$\frac{\Delta L}{L} = 1\% = \frac{1}{100}$$

$$\frac{\Delta V}{V} = \gamma \Delta T = 3\alpha \Delta T$$

$$\frac{\Delta L}{L} = \alpha \Delta T$$

$$\frac{\Delta L}{L} = \alpha \cdot 10$$

$$\frac{1}{100} = \alpha \cdot 10$$

$$\alpha = \frac{1}{100 \cdot 10} = \frac{1}{1000}$$

$$\frac{\Delta V}{V} = 3\alpha \cdot 10 = 3 \cdot \frac{1}{1000} \cdot 10$$

$$\frac{\Delta V}{V} \times 100\% = 0.03 \times 100\% = 3\%$$

3. The coefficient of linear expansion of brass and steel are α_1 and α_2 respectively. If we take a brass rod of length l_1 and length l_2 at 0°C , the difference in their lengths (l_1 and l_2) will remain the same at all temperatures, if
- (a) $\alpha_1 l_1 = \alpha_2 l_2$ (b) $\alpha_1 l_2 = \alpha_2 l_1$ (c) $\alpha_1^2 l_1 = \alpha_2^2 l_2$ (d) $\alpha_1 l_2^2 = \alpha_2^2 l_1$

Ans: (a) $\alpha_1 l_1 = \alpha_2 l_2$

$$\Delta l_1 = \Delta l_2$$

$$\alpha_1 l_1 \Delta T = \alpha_2 l_2 \Delta T$$

$$\alpha_1 l_1 = \alpha_2 l_2$$

4. Water falls from a height of 500 m. What is the rise in the temperature of water at the bottom, if the whole energy remains in water? (Specific heat of water = 4180 J/kg K, $g = 9.8 \text{ m/s}^2$)
 (a) 1.17 K (b) 0.24 K (c) 0.40 K (d) 0.19 K

Ans: (a) 1.17 K

Consider the case of 1kg of water.

$$E = mgh$$

where $m = 1\text{kg}$, $g = 9.8$, $h = 500\text{m}$ (given)

$$E = 1 \times 9.8 \times 500$$

$$E = 4900 \text{ J}$$

This is the energy stored in water as Q.

Specific heat of water is $c = 4180 \text{ J/(kg K)}$

$$Q = mc \Delta T$$

$$\text{Thus } \Delta T = Q/mc$$

$$\Delta T = 4900 / 1 \times 4180$$

$$\Delta T = 1.17 \text{ K}$$

5. 50 g of ice at 0°C is mixed with 5 g of water at 80°C , final temperature of the mixture will be ($L = 80 \text{ cal/g}$, $c = 1 \text{ cal/g } ^\circ\text{C}$)
 (a) 0°C (b) 40°C (c) 60°C (d) 4°C

Ans: (a) 0°C

Heat required to melt 50 g of ice = $mL = 50 \times 80 = 4000 \text{ cal}$

Heat given out by water in cooling from 80°C to $0^\circ\text{C} = m c \Delta T = 50 \times 1 \times 80 = 4000 \text{ cal}$

Heat given by water is just sufficient to melt the whole ice. So the final temperature is 0°C

6. A body of length 1m having a cross-sectional area 0.75 m^2 has heat flow through it at the rate of 6000 J/s. If $K = 200 \text{ Jm}^{-2}\text{K}^{-1}$, then the temperature difference across the ends of the body is
 (a) 20°C (b) 40°C (c) 80°C (d) 100°C

Ans: (b) 40°C

$$\frac{Q}{t} = \frac{KA\Delta T}{l}$$

$$\Rightarrow 6000 = \frac{200 \times 0.75 \times \Delta T}{1}$$

$$\therefore \Delta T = \frac{6000 \times 1}{200 \times 0.75} = 40^\circ\text{C}$$

7. If a body is heated from 27°C to 927°C , then the ratio of their energies of radiations emitted will be
 (a) 1 : 4 (b) 1 : 64 (c) 1 : 16 (d) 1 : 256

Ans: (d) 1 : 256

$$E = \sigma T^4$$

where σ is Stefan's constant.

$$\text{Given, } T_1 = 27^{\circ}\text{C} = 27 + 273 = 300\text{ K}$$

$$T_2 = 927^{\circ}\text{C} = 1200\text{ K}$$

$$\therefore \frac{E_1}{E_2} = \frac{(300)^4}{(1200)^4} = \frac{1}{256}$$

8. The temperatures of two stars are in the ratio 3 : 2. If the wavelengths of maximum intensity of the first body is $4000\text{ \AA}$, what is the corresponding wavelength of second body?
 (a) $9000\text{ \AA}$ (b) $6000\text{ \AA}$ (c) $2000\text{ \AA}$ (d) $8000\text{ \AA}$

Ans: (b) $6000\text{ \AA}$

According to Wien's displacement law

$$\lambda_m T = \text{constant}$$

$$\therefore \frac{(\lambda_m)_1}{(\lambda_m)_2} = \frac{T_2}{T_1}$$

$$\text{Here, } \frac{T_1}{T_2} = \frac{3}{2}, (\lambda_m)_1 = 4000\text{ \AA} = 4000 \times 10^{-10}\text{ m}$$

$$\therefore (\lambda_m)_2 = \frac{4000 \times 10^{-10} \times 3}{2}$$

$$= 6000 \times 10^{-10}\text{ m}$$

$$= 6000\text{ \AA}$$

9. On a cold morning, a metal surface will feel colder to touch than a wooden surface, because
 (a) metal has high specific heat (b) metal has high thermal conductivity
 (c) metal has low specific heat (d) metal has low thermal conductivity

Ans: (b) metal has high thermal conductivity

10. Consider two rods of the same length and different specific heats (c_1, c_2), thermal conductivities (K_1, K_2) and area of cross sections (A_1, A_2) and both having temperatures (T_1, T_2) at their ends. If their rate of loss of heat due to conduction is equal, then

- (a) $K_1 A_1 = K_2 A_2$ (b) $K_1 A_1 / c_1 = K_2 A_2 / c_2$ (c) $K_2 A_1 = K_1 A_2$ (d) $K_2 A_1 / c_2 = K_1 A_2 / c_1$

Ans: (a) $K_1 A_1 = K_2 A_2$

$$\left(\frac{dQ}{dt}\right)_1 = \left(\frac{dQ}{dt}\right)_2$$

$$\Rightarrow \frac{K_1 A_1 \Delta T_1}{l_1} = \frac{K_2 A_2 \Delta T_2}{l_2}$$

$$\therefore K_1 A_1 = K_2 A_2 \text{ [As } \Delta T_1 = \Delta T_2 = (T_1 - T_2) \text{ and } l_1 = l_2 \text{ given]}$$

For questions 11 to 15, two statements are given- one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to these questions from the options as given below.

- A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- C. Assertion is true but Reason is false.
- D. Both Assertion and Reason are false.

11. **Assertion:** In a pressure cooker the water is brought to boil. The cooker is then removed from the stove. Now on removing the lid of the pressure cooker, the water starts boiling again.

Reason: The impurities in water bring down its boiling point.

Ans: C

12. **Assertion:** For higher temperatures, the peak emission wavelength of a black body shifts to lower wavelengths.

Reason: Peak emission wavelength of a black body is proportional to the fourth power of temperature.

Ans: C

13. **Assertion:** When hot water is poured in a beaker of thick glass, the beaker cracks.

Reason: Outer surface of the beaker expands suddenly.

Ans: C

14. **Assertion:** A beaker is completely filled with water at 4°C. It will overflow, both when heated or cooled.

Reason: There is expansion of water below and above 4°C.

Ans: A

15. **Assertion:** It is hotter over the top of a fire than at the same distance of the side.

Reason: Air surrounding the fire conducts more heat upward.

Ans: C